

NEB Class 12 Management Math

Shivaji Chaulagain

August 9, 2025

Unit 2: Binomial Theorem

Binomial Theorem (Positive Integer Index)

- The binomial expansion of $(x + a)^n$ for $n \in \mathbb{N}$ is:

$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} a^k$$

- Each term in the expansion is called a **binomial term**.
- The general term (T_{k+1}) is:

$$T_{k+1} = \binom{n}{k} x^{n-k} a^k$$

- Example:** Expand $(x + 2)^4$:

$$\begin{aligned} &= \binom{4}{0} x^4 + \binom{4}{1} x^3 (2) + \binom{4}{2} x^2 (2)^2 + \binom{4}{3} x (2)^3 + \binom{4}{4} (2)^4 \\ &= x^4 + 8x^3 + 24x^2 + 32x + 16 \end{aligned}$$

Binomial Theorem Using nC_r Form

- Standard expansion:

$$(x + a)^n = {}^nC_0 x^n a^0 + {}^nC_1 x^{n-1} a^1 + {}^nC_2 x^{n-2} a^2 + \cdots + {}^nC_n x^0 a^n$$

- Example:** Expand $(x + 3)^3$:

$$\begin{aligned} &= {}^3C_0 x^3 + {}^3C_1 x^2 (3) + {}^3C_2 x (3)^2 + {}^3C_3 (3)^3 \\ &= x^3 + 3x^2 (3) + 3x (9) + 27 = x^3 + 9x^2 + 27x + 27 \end{aligned}$$

Finding the Middle Term

- If n is even, the middle term is:

$$T_{\frac{n}{2}+1}$$

- If n is odd, there are two middle terms:

$$T_{\frac{n+1}{2}} \text{ and } T_{\frac{n+3}{2}}$$

- Example 1:** Find the middle term of $(x + 1)^6$:

$$n = 6 \Rightarrow \text{middle term is } T_4 = {}^6C_3 x^3 = 20x^3$$

- Example 2:** Find the middle terms of $(x + 1)^5$:

$$n = 5 \Rightarrow \text{middle terms are } T_3 \text{ and } T_4 = {}^5C_2 x^3 + {}^5C_3 x^2 = 10x^3 + 10x^2$$

Finding the Independent Term

- The independent term is the term that does not contain x .
- Set the power of x to 0 in the general term and solve for k .
- **Example:** Find the independent term in $(x + \frac{1}{x})^6$:

$$T_{k+1} = {}^6C_k x^{6-k} \left(\frac{1}{x}\right)^k = {}^6C_k x^{6-k-k} = {}^6C_k x^{6-2k}$$

$$6 - 2k = 0 \Rightarrow k = 3 \Rightarrow T_4 = {}^6C_3 = 20$$

Finding the Coefficient of a Given Power of x

- Use the general term:

$$T_{k+1} = {}^nC_k x^{n-k} a^k$$

- Equate x 's exponent to required power and solve for k .
- **Example:** Find the coefficient of x^3 in $(x + 2)^5$:

$$T_{k+1} = {}^5C_k x^{5-k} (2)^k \Rightarrow 5 - k = 3 \Rightarrow k = 2$$

$$T_3 = {}^5C_2 x^3 (2)^2 = 10 \cdot 4 = 40 \Rightarrow \text{coefficient is } 40$$

Exponential Series

The exponential function e^x can be expanded using the infinite series:

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Important notes:

- The series is valid for all real and complex values of x .
- When x is negative, the series converges to e^{-x} :

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$$

Example 1: Expand e^1 up to 4 terms.

$$e^1 \approx 1 + \frac{1}{1!} + \frac{1^2}{2!} + \frac{1^3}{3!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} = \frac{8.5}{3} \approx 2.666$$

Example 2: Find the value of e^{-1} up to 4 terms.

$$e^{-1} \approx 1 - \frac{1}{1!} + \frac{1^2}{2!} - \frac{1^3}{3!} = 1 - 1 + 0.5 - 0.1667 = 0.3333$$

Logarithmic Series

The logarithmic function $\log(1+x)$ can be expanded as:

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

This expansion is valid for $-1 < x \leq 1$.

Special Cases:

- $\log(1+x)$ is valid for $-1 < x \leq 1$
- $\log(1-x)$ can be derived by replacing x with $-x$ in the expansion of $\log(1+x)$:

$$\log(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots\right) = -\sum_{n=1}^{\infty} \frac{x^n}{n}, \quad |x| < 1$$

Example 1: Expand $\log(1.1)$ up to 3 terms. Let $x = 0.1$

$$\log(1.1) \approx 0.1 - \frac{0.1^2}{2} + \frac{0.1^3}{3} = 0.1 - 0.005 + 0.000333 = 0.0953$$

Example 2: Expand $\log(0.9)$ up to 4 terms. Let $x = -0.1$

$$\log(0.9) = \log(1-0.1) \approx -0.1 - \frac{(0.1)^2}{2} - \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} = -0.1 - 0.005 - 0.000333 - 0.000025 = -0.10536$$

Example 3: Expand $\log(1-x)$ up to 3 terms where $x = 0.2$

$$\log(1-0.2) \approx -0.2 - \frac{(0.2)^2}{2} - \frac{(0.2)^3}{3} = -0.2 - 0.02 - 0.002667 = -0.2227$$

Note: These expansions are useful for approximations and theoretical derivations in calculus and algebra.

Unit 2: Binomial Theorem

Short Answer Questions

1. Expand $(2x - \frac{1}{x})^5$ using the Binomial Theorem.
2. Find the coefficient of x^3 in the expansion of $(x+2)^6$.
3. Find the middle term in the expansion of $(x^2 - \frac{3}{x})^8$.
4. In the expansion of $(1+3x)^7$, find the term independent of x .
5. Find the term containing x^5 in the expansion of $(x^2 + \frac{1}{x})^7$.

Long Answer Questions

6. Prove that in the expansion of $(1+x)^n$, the sum of the coefficients is 2^n .
7. Find the coefficient of x^6 in the expansion of $(2x - \frac{3}{x^2})^9$.
8. Show that the middle term of $(1+x)^{2n}$ is $\binom{2n}{n}x^n$. Verify your result for $n = 4$.
9. Find the general term of the expansion $(a+bx)^n$ and hence find the coefficient of x^k in $(3-2x)^8$.
10. If the 5th and 6th terms in the expansion of $(1+x)^n$ are equal, find the value of n .